

GURLEEN KAUR

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EDUCATION

University of Auckland — BSc, Computer Science & Biology

2022 – Present

EXPERIENCE

Software Engineering Intern — Scopious, Auckland NZ

Nov 2025 – Feb 2026

- Owned a GPS asset-tracking feature end to end — requirements, design, backend, frontend, testing — and shipped it to production, where it tracks **50+ IoT devices** over a LoRaWAN backend and ships as a documented product capability.
- Eliminated redundant re-renders on the Live Map with an update queue over the polling layer, keeping it responsive as device count and polling frequency scaled; added marker clustering, per-device colouring, and device toggling.
- Built Historic Playback (per-device trails, hover-to-focus dimming, 4-device concurrency cap) loading timeline data only for the focused device, plus fleet scoping by pool, tag, device, or org.

Full-Stack Developer — OnChain, Web Development & Consulting Club (WDCC)

Mar 2026 – Present

- Expanding OnChain, WDCC's flagship product, from a chat forum into a DAO discussion space (Next.js, React, TypeScript, Supabase) ahead of a live platform launch.
- Built comment CRUD REST endpoints, backend-backed read/unread state, and image uploads for a live client — an edtech startup expanding across NZ, Australia, and Asia — in Agile sprints with direct client feedback; added auth-gated access and soft-delete moderation with admin override to preserve audit trails.
- Diagnosed and fixed the forum's data layer: killed a 401 flicker and duplicate fetches by gating requests on resolved auth, then added a shared TanStack Query cache keyed by DAO, post, and sort, removing the full reload on every Forum tab switch and keeping content visible during revalidation.

Teaching Assistant, COMPSCI 101 (Python) — University of Auckland

Jul 2025 – Jul 2026

- Mentor **100+ students** weekly in Python fundamentals, debugging, and logic tracing across labs and code review.

Data Analysis Intern — Insurance Division, PwC

Jan 2024 – Feb 2024

- Automated ETL pipelines (Python, pandas) over **10,000+ records**, replacing a manual process; built Excel risk dashboards for consulting stakeholders.

PROJECTS

Scaffold — AI-Native Learning Infrastructure for Universities · Co-Founder

2026 – Present

TypeScript, React, Node.js, VS Code Extension API, MCP, SVG

- Selected for Velocity Innovation Challenge Stage 2 — top 30 of 150+ startups**; won the WDCC × SESA Hackathon 2026 with the prototype.
- Building a VS Code extension that keeps CS students thinking while an AI agent writes their code: it streams the agent's actions, parks the build at consequential architecture decisions (auth model, data storage) for the student to make, and logs the reasoning to a `LEARNING_LOG.md` they submit with the assignment.
- Architected the MVP as one TypeScript process — an MCP-over-HTTP server in the extension host that any MCP-speaking agent calls at a fork, plus a session-transcript tailer driving the live panel — cutting the prototype's Python runtime, API key, and shell hooks, so install is zero-config and Scaffold is never in the agent's tool path.
- Running customer discovery with University of Auckland CS faculty and students, advised by three Professional Teaching Fellows on rollout, procurement, and academic-integrity workflows, with classroom pilot testing next; built the educator insight dashboard in hand-rolled responsive SVG, surfacing cohort concept gaps without surveilling individuals.

Erica — AI Marketing Intelligence Platform · Capstone, Tech Lead of 5 engineers

[GitHub](#) | [Case Study](#)

Python, FastAPI, AWS Bedrock, React / Next.js, PostgreSQL, Docker

- Owned the system architecture for a multi-agent AI platform that detects ad fatigue in real time and recommends fixes; replaced a rigid LangGraph pipeline with a **16-tool tool-calling agent** on AWS Bedrock Converse, making the flow conversational and multi-step instead of fixed routing; built a two-tier recommendation engine where one LLM reads the ad creative visually and a second grounds the fix in live research.
- Designed the LLM security layer — prompt-injection and jailbreak guards, rate limiting, PII scrubbing — validated against a **190+ prompt OWASP-aligned adversarial suite**.

Agribuddy — AI Pre-Visit Briefing Tool · Winner, BNZ × AWS AI Innovate Hackathon 2026

[GitHub](#) | [Demo](#)

Python, FastAPI, AWS Bedrock (Converse API), Bedrock Knowledge Bases, S3, Next.js

- Validated the problem with BNZ agribusiness advisors — hours of manual pre-visit prep, then no connectivity on-farm — and built the backend AI layer as a FastAPI service exposing a shared agent "brain" to multiple clients.
- Designed a multi-agent orchestration system: a supervisor plus five specialists (farm financials, market intelligence, credit, compliance, relationship), with deterministic ratio and lending-framework logic kept in code outside the LLM, so every number is calculated rather than generated.
- Built retrieval-augmented context over an industry-tagged Bedrock Knowledge Base on an S3 corpus with per-document metadata sidecars, assembling ratios, flags, market intel, ranked questions, and inline citations behind one endpoint.

AWARDS & LEADERSHIP

- Winner** — BNZ × AWS AI Innovate Hackathon 2026 (Agribuddy), WDCC × SESA Hackathon 2026 (Scaffold), Partly Hackathon 2025. **2nd** — Fonterra AI Agents Challenge 2025 (Fabric NL-to-SQL agent). **Most Creative** — Partly 2026. **Best Presenter** — Trade Me Thinkathon 2025. **\$560 Spark scholarship**, We The Women Hackfest, from thousands of applicants.
- Technical Executive, Google Developer Groups on Campus** (Mar 2026 – Present) — deliver web, AI/ML, and cloud workshops to 100+ students; coordinate hackathons and judging for 50+ teams; Lead Organiser, Women in Tech panel.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, SQL, HTML, CSS

Web & Backend: React, Next.js, SSR, Node.js, Express, FastAPI, REST APIs, SSE, TanStack Query, SVG data visualisation

Data & Cloud: PostgreSQL, Supabase, MongoDB, Firebase, AWS (EC2, S3, Bedrock, Knowledge Bases), Docker, pandas

Tools & AI: Git, GitHub Actions, CI/CD, Pytest, Agile/Scrum, LLM tool-calling, RAG, MCP, LangChain, OpenCV